

JAYASHANKA ANUSHAN

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PROFESSIONAL SUMMARY

Robotics Enthusiast with hands-on experience in ROS/ROS2-based autonomous systems, sensor integration, embedded platforms, and robot control applications. Skilled in Python, C++, Linux, Docker, Gazebo simulation, and hardware-software integration. Experienced in developing robotic systems using LiDAR, cameras, IMU, Arduino, ESP32, and industrial robot platforms. Currently pursuing MSc Robotics with practical experience in autonomous navigation, robot communication, system integration, and real-time robotic systems.

ROBOTICS & AUTONOMOUS SYSTEMS PROJECTS

- **Smart Cart with ROS2 Jazzy (Group Project) | 2025**

Developed an autonomous mobile robot simulation using ROS2 Jazzy and Gazebo. Integrated LiDAR-based obstacle detection, UWB positioning, and Kalman filtering for user-following behaviour. Worked on navigation logic, sensor integration, robot communication, and multi-package ROS system integration for autonomous operation.

- **CARA – Companion Assistive Robot for the Aged | Ongoing | 2026**

Designed and under develop an assistive mobile robot concept focused on elderly monitor and support. Integrated ROS-based architecture, embedded controllers, sensors, and AI-assisted interaction features. Under development of robot simulation, control systems, and human-robot interaction components.

- **Rose v1.0 (Humanoid Robot Face) | 2025**

Developed an interactive humanoid robotic face using Arduino, ROS, and computer vision. Designed mechanical components using SolidWorks and controlled servo-based facial movements for human interaction. Implemented face recognition using MediaPipe and ROS communication for real-time robotic behaviour.

- **RoboDK Program with Epson VT6 | 2025**

Programmed Epson VT6 industrial robot using Epson RC+ 7.0 for pick-and-place, drawing, and object manipulation tasks. Developed automation workflows using Python scripting and industrial robot programming techniques.

- **RobotSC (Robot Statics/ Control) v1.0 | 2025**

Developed a distributed robot monitoring and control framework inspired by ROS communication architecture. Implemented WebSocket-based real-time communication, FastAPI backend services, and React-based monitoring interface for robot data visualization and control.

- **JS_Blind Talk v1.0 | 2025**

Designed an Arduino-based assistive communication system using embedded electronics and real-time input processing. Developed a human interaction interface for converting user input into readable and audible communication.

ROBOTICS TOOLS & ENVIRONMENTS

- **ROBOTICS SOFTWARE:**

- ROS2 Jazzy, ROS Noetic, Gazebo, RViz, Docker, Git, VS Code

- **PROGRAMMING:**

- Python, C++, C, FastAPI

- **EMBEDDED & HARDWARE:**

- Arduino, ESP32, Raspberry Pi, LiDAR, IMU, Cameras, Servo Motors

EDUCATION

- **Middlesex University Dubai (2024 - 2026)**
MSc Robotics
 - **London Metropolitan University (2019 - 2021)**
BEng (Hons) Software Engineering with First Class Honours
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PROFESSIONAL EXPERIENCE

Wayamba University of Sri Lanka
IT and Innovation Technical Officer

September 2019 - March 2023

- Maintained and troubleshoot 50+ laboratory equipment systems across four innovation laboratories, including automation systems, electrical trainer panels, and IT infrastructure.
- Assisted with setup and configuration of educational lab tools (e.g., **PLC, Proteus**).
- Supported technical operations and logistics for large-scale university events.
- Instructed over **500+** students regarding academic practical and training sessions

OPPEIN, UAE
Operations Coordinator

September 2024 - Present

- Supported technical operations through process optimization and digital workflow improvements.
- Automated reporting and data management processes using Excel and enterprise systems.
- Coordinated technical teams, documentation, and operational activities.

Alfred TALKE Logistic Services Qatar LLC.
Inventory Administrator

April 2023 - July 2024

- Managed technical inventory operations and system-based tracking processes.
- Improved operational accuracy through data analysis and digital tools.
- Supported large-scale logistics technology workflows.